
Auction-Based Online Policy Adaptation for Evolving Objectives

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We consider multi-objective reinforcement learning problems where objectives
2 come from an identical family—such as the class of reachability objectives—and
3 may appear or disappear at runtime. Our goal is to design adaptive policies that
4 can efficiently adjust their behaviors as the set of active objectives changes. To
5 solve this problem, we propose a modular framework where each objective is
6 supported by a selfish local policy, and coordination is achieved through a novel
7 *auction*-based mechanism: policies bid for the right to execute their actions, with
8 bids reflecting the urgency of the current state. The highest bidder selects the
9 action, enabling a dynamic and interpretable trade-off among objectives. Going
10 back to the original adaptation problem, when objectives change, the system adapts
11 by simply adding or removing the corresponding policies. Moreover, as objectives
12 arise from the same family, identical copies of a parameterized policy can be
13 deployed, facilitating immediate adaptation at runtime. We show how the selfish
14 local policies can be computed by turning the problem into a general-sum game,
15 where the policies compete against each other to fulfill their own objectives. To
16 succeed, each policy must not only optimize its own objective, but also reason
17 about the presence of other goals and learn to produce calibrated bids that reflect
18 relative priority. In our implementation, the policies are trained concurrently using
19 proximal policy optimization (PPO). We evaluate on Atari Assault and a gridworld-
20 based path-planning task with dynamic targets. Our method achieves substantially
21 better performance than monolithic policies trained with PPO.

22 1 Introduction

23 Consider the problem of controlling a mobile robot deployed in a
24 cat shelter whose task is to deliver food to cats roaming around
25 the facility (Figure ??). New food requests from hungry cats may
26 arrive at arbitrary times and locations, while existing requests may
27 disappear if a cat wanders away before being served. Since the robot
28 can serve only one cat at a time, fulfilling one request may delay
29 others, potentially allowing some cats to move farther away. Thus,
30 the robot must dynamically adapt to the set of active requests to
31 maximize the number of cats fed before they disappear.

32 In this work, we model the above control problem as a multi-
33 objective reinforcement learning task in which all the objectives
34 come from an identical family, such as reachability in this case.
35 These objectives can be conflicting and any number of them may be active at a given time. Moreover,
36 we make no assumptions about their distribution or arrival pattern. Our goal is to design adaptive

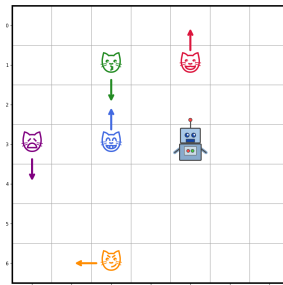


Figure 1: Cat Feeder env.

37 policies that update their behavior seamlessly as objectives appear or disappear. This is a natural
38 problem relevant to many real-world scenarios and has been extensively studied in the planning
39 literature (?), typically assuming that a precise environment model (like a graph) is available and there
40 is no uncertainties. In contrast, within reinforcement learning (RL) settings—where the environment
41 may be uncertain and a model may not be available—this problem has not been studied to the best of
42 our knowledge.

43 To solve this problem, we propose a compositional RL framework, where each objective is served
44 using a local policy, selfishly maximizing its assigned objective by competing against the other
45 policies. Coordination between these policies is achieved through a novel auction-based mechanism:
46 at every decision step, policies submit bids for the right to execute their actions, where bids reflect
47 the urgency of the current state with respect to their objectives. The highest bidder selects the action,
48 enabling a dynamic and interpretable trade-off among objectives. Returning to the cat feeder example,
49 suppose the robot approaches an intersection with two active requests, the policy for the first request
50 recommends turning left, while the policy for the second recommends turning right. Since both
51 actions cannot be executed simultaneously, a trade-off is unavoidable. The auction mechanism
52 determines which objective is prioritized, based on the relative urgency encoded in the bids.

53 This compositional design yields strong modularity, facilitating fast adaptation: when an objective
54 appears or disappears, only the corresponding policy needs to be added or removed. Furthermore,
55 when objectives belong to the identical family—which is reachability in our case—a single generalized
56 policy can be trained for the entire family. We can deploy identical *copies* of this policy when new
57 objectives arrive, enabling immediate adaptation. Modularity also permits us to define objective-
58 specific reward signals while the auction mechanism takes care of the trade-offs between objectives.
59 For example, we can easily define a distance-based dense reward signal for each policy controlling
60 the cat feeder to move closer to its assigned cat. In contrast, it is challenging to define such a dense
61 reward signal to train a monolithic policy to trade-off between the multiple objectives. We have to
62 resort to imposing a preference ordering over the multi-dimensional reward vector, or stick to sparse
63 rewards. We show in our experiments that monolithic policies, trained with both sparse rewards and
64 different types of scalarized dense rewards, fall much short of the payoffs achieved by our modular
65 approach.

66 Yet another advantage of our modular framework is interpretability. At any moment, we can determine
67 the policy in control, and thereby the objective currently being pursued by the agent. This explanation
68 is beneficial in a range of real-world situations, like providing post-hoc explanations of failures, and
69 also feedback on biases in pursuing different objectives.

70 We implement our modular framework as a general-sum game between agents that control the local
71 policies to fulfill their assigned objectives, and this game is set up using a simple wrapper around
72 existing multi-objective environments. Each agent’s action space is augmented with an additional
73 numeric variable representing the bids, and the transitions between the game’s states encode the
74 execution of the action selected by the highest bidder. However, we still need to address some
75 challenges.

76 **Challenge I: Enforcing honest bids.** While the auction mechanism treats all policies fairly, the
77 policies must still place honest bid in proportion to their urgency. Otherwise, they may learn to overbid
78 and suppress competitors, resulting in suboptimal results. To prevent this, we impose penalties during
79 training that scale with the bid values. As a consequence, excessive bidding reduces the net return
80 of a policy, thus aligning its incentives with truthful urgency estimation—a policy only bids high
81 when it is sure to receive a reward (when it feeds the cat) or to avoid a penalty (when a cat is close to
82 disappearing).

83 **Challenge II: Achieving environment awareness.** Effective bidding depends not only on the current
84 state and the local objective, but also on the configuration of competing objectives. For instance, in
85 the cat feeder problem, if two requests are spatially close, their policies need not compete aggressively.
86 Conversely, if requests lie in opposite directions, the urgent policy must bid sufficiently high to win.
87 To enable such environment-aware bidding, we allow the policies to observe all the objectives, i.e.,
88 the policies have information about the global state. Additionally, the policies are trained concurrently
89 so that they learn to optimize payoffs in the presence of opponents.

90 **Challenge III: Unbounded objective count.** When policies have to compete with many objectives,
91 providing each with the global state results in a very high dimensional observation space. This

92 makes training slow with a standard neural network policy. We address this with an attention pooling
 93 mechanism that aggregates information about an arbitrary number of competing objectives into a
 94 fixed-dimensional representation. Each local policy is equipped with its own pooling module which
 95 is co-trained with the policy. A bonus of this architecture is that it allows policies to compete with
 96 more objectives than those during training, since the pooling mechanism can extract useful features
 97 from an arbitrary number of objective states.

98 We test our framework on two multi-objective environments: the cat feeder environment implemented
 99 as a gridworld, and the Atari Assault game from the Arcade Learning Environment (?). To train our
 100 policies, we take inspiration from ? who provide guidance on using proximal policy optimization
 101 (PPO; ?) in multi-agent systems. We observe empirical convergence of training and substantially
 102 better performance compared to baselines in both environments. Lastly, to aid interpretability in
 103 renderings of rollouts, we implement visualization of the policy in control and the objective being
 104 pursued.

105 Related Work

106 A large body of work in multi-objective reinforcement learning (MORL) relies on *scalarization*,
 107 where multiple reward functions are aggregated into a single scalar objective so that standard RL
 108 algorithms can be applied. The simplest approach uses weighted sums of rewards (?), though
 109 nonlinear scalarization methods have also been proposed (?). A key limitation is that the induced
 110 importance among rewards may not reflect the designer’s intent, often leading to a tedious debugging
 111 process (?). In contrast, our approach achieves a trade-off among reward components without
 112 collapsing them into a fixed scalar objective.

113 Other works address trade-offs by adopting a specific optimality notion, such as Pareto optimality (?)
 114 and its approximations (?), or fairness-based criteria across rewards (???). These approaches typically
 115 learn a single monolithic policy satisfying the chosen criterion. By contrast, we learn independent,
 116 selfish policies for each reward component and compose them at runtime, preserving modularity
 117 while achieving a coherent global trade-off.

118 Only a few works consider distributed policies for multiple rewards. W-learning (?) and its deep RL
 119 extension (?) train separate policies alongside meta-functions that score states by urgency, selecting
 120 the policy with the highest score at runtime. Other methods aggregate decisions through ranked
 121 voting (?) or fixed rules such as summing action values (?). Our approach is simpler: it uses an
 122 engineered reward structure that enables standard RL algorithms (e.g., PPO) without additional
 123 meta-policies or complex aggregation. Moreover, to the best of our knowledge, we are the first to
 124 study the incremental MORL setting, where reward components can be added or removed at runtime.

125 The idea of bidding-based selfish policies is inspired by auction mechanisms for multi-objective
 126 path planning on graphs (?) and by the literature on bidding games (???). These works use explicit
 127 model-based approaches and do not readily extend to the model-free setting of RL. Besides, they are
 128 restricted to the setting with two objectives.

129 2 Multi-Objective Discounted Reachability with Deadlines

130 We formalize the policy synthesis problem for a fixed number of goals, and then introduce our
 131 auction-based framework for this setting. Afterwards, we will describe how our framework naturally
 132 extends to the setting with dynamic goal appearance and disappearance.

133 **Policy synthesis for a fixed goal set.** We use multi-objective Markov decision processes (MO-MDP)
 134 to model uncertain environments with a *fixed* number of discounted reachability objectives with dead-
 135 lines. An MO-MDP with $m \in \mathbb{N}_{>0}$ objectives is specified by a tuple $\mathcal{M} = (S, A, T, D, R, \gamma, M, \mu)$
 136 where S is the set of *locations* equipped with the Euclidean metric $d(\cdot, \cdot)$, with $\mathbf{S} = S^{m+1}$ being
 137 the *state space* such that $(s_0, s_1, \dots, s_m) \in \mathbf{S}$ includes the agent’s location s_0 and the location of
 138 the i -th goal for every i , A is the set of *actions*, $T : \mathbf{S} \times A \rightarrow \Delta(\mathbf{S})$ is the probabilistic *transition*
 139 *function*, $D = \{d_i \in \mathbb{N} \cup \infty\}_{i \in [1;m]}$ is the set of goal-specific *deadlines* ($d_i = \infty$ effectively
 140 means the i -th goal has no deadline), and without loss of generality, we assume $d_1 \leq \dots \leq d_m$,
 141 $R = \{r_i \in \mathbb{R}_{>0}\}_{i \in [1;m]}$ is the set of *rewards* obtained after the completion of each goal, $\gamma \in (0, 1)$ is
 142 the *discount factor*, $M > r_{\max} := \max_{i \in [1;m]} r_i$ is the per-objective one-time *penalty* for missing the
 143 deadline, **KM: the penalty is always added undiscounted** and $\mu \in \mathbb{D}(\mathbf{S})$ is the *initial state distribution*.

A policy in an MO-MDP $\mathcal{M}$ is a function $\pi : (\mathbf{S} \times A)^* \times \mathbf{S} \rightarrow \mathbb{D}(A)$ that maps a history of state-action pairs and the current state to a distribution over actions. **KM: Restrict to memoryless policies?** The policy π is executed until the last deadline d_m , giving rise to a random sequence of states and actions $\rho = s^0, a^0, s^1, a^1, \dots, s^{d_m}, a^{d_m}$, called a *trace*. For each $i \in [1; m]$, let τ_i be the first time point at which the i -th goal is reached, i.e., $s_{0^i}^{\tau_i} = s_i^{\tau_i}$, and $\tau_i = \infty$ if the i -th goal is never reached. The per-objective reward collected on the trace ρ is $w_i(\rho) = \gamma^{\tau_i} r_i$ if $\tau_i < d_i$, and $w_i(\rho) = -M$ otherwise. The total reward collected on ρ is: $w(\rho) = \sum_{i=1}^m w_i(\rho)$. The value of the policy π , denoted V^π , is given as $V^\pi = \mathbb{E}_\rho^\pi[w(\rho)]$. Our baseline is the *optimal policy* π^* that maximizes the value, i.e., $\pi^* = \arg \max_\pi V^\pi$, and the *optimal value* is V^{π^*} .

Lower complexity bound: simplified setting with deterministic agent and stationary goals. We establish that the optimal policy synthesis problem is NP-hard, and we show this already on a simplified setting: Suppose the transition function T of the MO-MDP is deterministic, the goals are stationary such that the location of the i -th goal is always $g_i \in S$, and for any policy, the minimum travel time from a given location $x \in S$ to the i -th goal is $t_i = d(x, g_i)$. (Alternatively, t_i can be interpreted as the length of the shortest path from x to g_i when the agent moves at the unit velocity.)

Theorem 1. *Finding the optimal policy for the MO-MDP is NP-hard.*

Proof. The simpler setting with deterministic agents and stationary goals subsumes the Euclidean traveling salesman problem, where the deadlines are all infinity, and the goal is to find the shortest path passing through all the goals. This problem is known to be NP-complete ?, and our deadline-restricted case is at least as hard as it. **KM: There is a slight difference, that we allow revisiting a state. However, I found on discussion forums that this does not change the complexity, find a reference. Also see metric traveling salesman, which is more general (we could then replace the Euclidean distance requirement in the MO-MDP state space with the requirement of just having a metric. □**

Proposition 1. *Consider the simpler case of deterministic agent with stationary goals. There exists a permutation $\sigma^* \in \Sigma_m$, where Σ_m is the set of all permutations of $[1; m]$, such that the optimal policy π^* pursues the goals in the order $\sigma^*(1) \dots \sigma^*(m)$, and π^* takes the shortest paths from the initial agent location to goal $\sigma^*(1)$ and between $\sigma^*(i)$ and $\sigma^*(i+1)$, for every $i \in [1; m-1]$.*

The above proposition effectively reduces the problem to an optimal *scheduling* problem. Furthermore, Theorem ?? suggests that a polynomial-time centralized optimal algorithm does not exist, unless $P=NP$, already for the simpler deterministic case with stationary goals. This motivates us to seek for a heuristic approach that is aimed towards finding the optimal schedule over the goals whose existence is guaranteed by Proposition ??.

A natural deterministic baseline is the *least slack time first* (LSTF) rule, which schedules the goal with smallest slack $slack_i(x, t) := d_i - t - d(x, g_i)$. However, this notion of urgency is tailored to stationary goals and deterministic travel times, and does not align with our full MO-MDP objective once targets move stochastically or may appear and disappear over time. For this reason, we use a more general benchmark that we call *reward-aware urgency first* (RAUF). **KM: Is this used in the literature? If yes, some citation would be nice.** At every decision point, RAUF selects one of the goals with maximal *reward-aware urgency*, namely the marginal continuation-value gain from granting that goal immediate control rather than deferring control to competing goals; ties are broken uniformly at random. Unlike slack, this urgency notion directly accounts for discounted rewards, missed-deadline penalties, stochastic target motion, and future target arrivals or removals. Our auction mechanism is designed to approximate this reward-aware urgency ordering in a decentralized manner while remaining tractable in settings where computing the globally optimal schedule is infeasible. **GS: Also talk about how urgency encodes agent cooperation.**

3 Auction-Based Multi-Policy Framework

We consider the compositional approach to policy synthesis for MO-MDPs, where we will design a selfish, *local* policy maximizing each individual value component, towards the fulfillment of some required global coordination requirements. The main crux is in the composition process, where each local policy may propose a different action, but the composition must decide one of the actions that will be actually executed. Importantly, the composition must be implementable in a distributed

manner, meaning we will *not* use any global policy that would pick an action by analyzing all local policies and their reward functions.

We present a novel *auction*-based RL framework for compositional policy synthesis for MO-MDPs. In our framework, not only do the local policies emit actions, but they also *bid* for the privilege of executing their actions for a given number of time steps $\tau \in \mathbb{Z}_{>0}$ in future. The bids are all non-negative integers bounded by a given constant $\beta \in \mathbb{Z}_{\geq 0}$, and the highest bidder’s actions get executed for the following τ consecutive steps, with bidding ties being resolved uniformly at random. To discourage policies from always bidding high, we penalize the policies proportional to their bids scaled with a constant $\rho \in (0, 1)$. We consider two intuitive alternatives for the penalty model, namely Winner-Pays and All-Pay: in the Winner-Pays setting, only the executed policy (one of the highest bidders) pays the penalty, whereas in the All-Pay setting, all policies pay penalties. Through this mechanism, policies select their bids to reflect how critical their action is in the current state, and the bidding penalty incentivizes truthful urgency. This way, we obtain a purely decentralized scheme to coordinate local policies in a given MO-MDP. **GS: Maybe talk about existence of Nash equilibria here.**

The parameters τ , β , and ρ denote the *bidding interval*, *maximum bid*, and *bid penalty coefficient*, respectively. Ablation studies in Section ?? highlight their importance. If τ is too small, policies switch too frequently, risking suboptimal outcomes. If β is too small, policies cannot express preference strength, leading to near-random control. If ρ is too large, policies tend to abstain from bidding.

Theoretical analysis of the bidding schemes. We now show that both Winner-Pays and All-Pay implement the reward-aware urgency ordering whenever one goal has a sufficiently large urgency advantage over the others. To capture stochastic motion of goals, we formulate the urgency scores using conditional expectations under the transition kernel of the MO-MDP. The deterministic stationary-goal case discussed above is then recovered as a special case in which these conditional expectations collapse to point values. **KM: No need to talk about the deterministic case here.**

Suppose π_i is the local policy for the i -th goal, mapping every agent state $s \in S$ and time t to a bid b_i and an action a_i . For convenience, we will write $\pi_i^b: (s, t) \mapsto b_i$ and $\pi_i^a: (s, t) \mapsto a_i$ as two separate components of the policy. At a bidding round (s, t) , let $\Xi_i^{\text{win}}(s, t)$ and $\Xi_i^{\text{lose}}(s, t)$ denote the random future trajectories **KM: for unbounded time in the future?** induced by the environment and by the subsequent bids when objective i wins and loses the current auction, respectively. We first define the pre-payment continuation values

$$\begin{aligned}\tilde{V}_i^{\text{win}}(s, t) &:= \mathbb{E}[w_i(\Xi_i^{\text{win}}(s, t))] , \\ \tilde{V}_i^{\text{lose}}(s, t) &:= \mathbb{E}[w_i(\Xi_i^{\text{lose}}(s, t))] ,\end{aligned}$$

KM: Typo: s, t, i wins at (s, t) should be i wins at (s, t) . KM: I do not understand the notation $w_i(\Xi_i^{\text{win}}(s, t))$. where the expectation ranges over stochastic state transitions, target motion, target arrivals or removals, and future randomized bids and tie-breaking. KM: random arrival and removal of objectives is problematic, because this would cause the dimension of the state space to change, meaning the dimension of the underlying random variable (corresponding to “ $\mathbb{E}$ ” to change. I would keep the number of goals fixed, this is a distraction and unnecessarily complicates the math. The post-payment values when objective i bids b_i are

$$\begin{aligned}V_i^{\text{win}}(s, t, b_i) &:= \tilde{V}_i^{\text{win}}(s, t) - \rho b_i, \\ V_i^{\text{lose}}(s, t, b_i) &:= \tilde{V}_i^{\text{lose}}(s, t) - \rho b_i \cdot \mathbf{1}[\text{All-Pay}].\end{aligned}$$

The continuation value at a bidding state (s, t) is then

$$\forall i. V_i^{\text{cont}}(s, t) := \max_{b_i \geq 0} \mathbb{E}_{b_{-i} \sim \pi_{-i}^b(s, t)} [U_i(s, t, b_i, b_{-i})],$$

KM: There is something wrong: $V_{\text{win}}, V_{\text{lose}}, \tilde{V}_{\text{win}}, \tilde{V}_{\text{lose}}$ do not depend on V_i^{cont} . where $b_{-i} \sim \pi_{-i}^b(s, t)$ denotes the collection of bids of all policies other than i **KM: I think we need to say “...at the Nash equilibrium, ...”, and**

$$U_i(s, t, b_i, b_{-i}) := \begin{cases} V_i^{\text{win}}(s, t, b_i) & \text{if } b_i > \max_{j \neq i} b_j, \\ V_i^{\text{lose}}(s, t, b_i) & \text{if } b_i < \max_{j \neq i} b_j, \\ \frac{1}{|W|} V_i^{\text{win}}(s, t, b_i) + \left(1 - \frac{1}{|W|}\right) V_i^{\text{lose}}(s, t, b_i) & \text{if } i \in W, \end{cases}$$

with $W := \{k \in [1; m] \mid b_k = \max_{\ell \in [1; m]} b_\ell\}$ being the set of highest bidders. The *net gain* for bidding b_i at (s, t) is defined as:

$$\begin{aligned}\Delta_i^{\text{net}}(s, t, b_i) &:= V_i^{\text{win}}(s, t, b_i) - V_i^{\text{lose}}(s, t, b_i) \\ &= \Delta_i^{\text{gross}}(s, t) - \rho b_i \cdot \mathbf{1}[\text{Winner-Pays}],\end{aligned}$$

where the *gross gain* is

$$\Delta_i^{\text{gross}}(s, t) := \tilde{V}_i^{\text{win}}(s, t) - \tilde{V}_i^{\text{lose}}(s, t).$$

The quantity $\Delta_i^{\text{gross}}(s, t)$ is exactly the reward-aware urgency of policy i : it measures the continuation-value loss incurred by not allocating the next bidding interval to i . Thus RAUF selects a policy maximizing $\Delta_i^{\text{gross}}(s, t)$.

The following theorem establishes that the Winner-Pays setting implements RAUF whenever the leading urgency exceeds the runner-up by more than one bid increment.

Theorem 2. *In the Winner-Pays setting, consider a single bidding round at the agent state $s \in S$ and time t . Suppose $\rho\beta > \max_{j \in [1; m]} \Delta_j^{\text{gross}}(s, t)$. **KM:** *I would make it $\rho\beta > r_{\max} + M$. It is always better to impose constraints on the hyper-parameters, because maintaining those constraints is in the designer’s hand. However, a constraint on the Δ_j^{gross} feels like a matter of “assumption” or “luck.” and let i^* be a policy for which the following gross gain gap condition holds:**

$$\Delta_{i^*}^{\text{gross}}(s, t) - \max_{j \neq i^*} \Delta_j^{\text{gross}}(s, t) > \rho. \quad (1)$$

Then in any pure Nash equilibrium, policy i^* wins control.

Proof. Suppose for contradiction, some other policy $j \neq i^*$ wins control by bidding b_j , while policy i^* bids $b_{i^*} < b_j$. As this is a Nash equilibrium, it must hold that $\Delta_j^{\text{net}}(s, t, b_j) > 0$, i.e., $\Delta_j^{\text{gross}}(s, t) - \rho b_j > 0$, because otherwise, policy j would be better off by bidding zero to lose the bidding. This gives us the upper bound $\rho b_j < \Delta_j^{\text{gross}}(s, t)$. It follows from $\rho\beta > \max_{k \in [1; m]} \Delta_k^{\text{gross}}(s, t)$ that $\rho b_j < \rho\beta$, and hence $b_j < \beta$, i.e., the policy j does not make the highest possible bid.

Now we assumed that policy i^* had bid $b_{i^*} < b_j$, but we show that policy i^* would achieve a higher net gain by bidding $b_j + 1$, contradicting the Nash equilibrium: $\Delta_{i^*}^{\text{net}}(s, t, b_j + 1) = \Delta_{i^*}^{\text{gross}}(s, t) - \rho(b_j + 1) = \Delta_{i^*}^{\text{gross}}(s, t) - \rho b_j - \rho > \Delta_{i^*}^{\text{gross}}(s, t) - \Delta_j^{\text{gross}}(s, t) - \rho > 0$. Here, notice that $b_j + 1$ is a valid bid since $b_j < \beta$, as we established above. $\square$

The previous theorem is abstract in the sense that it only assumes an urgency gap. The following lemma shows how the missed-deadline penalty $-M$ induces such a gap when delaying a target materially increases its probability of expiring.

Lemma 1 (Expiry-sensitive urgency gap). *Fix a bidding state (s, t) and a policy i^* . Suppose there exists $\delta > 0$ such that **KM:** *I think such a δ would always exist. We need to say “for every $\delta > \dots$ ”**

$$\Pr[\tau_{i^*} = \infty \text{ or } \tau_{i^*} \geq d_{i^*} \mid s, t, i^* \text{ loses at } (s, t)] - \Pr[\tau_{i^*} = \infty \text{ or } \tau_{i^*} \geq d_{i^*} \mid s, t, i^* \text{ wins at } (s, t)] \geq \delta,$$

KM: *i^* loses at (s, t) **KM:** *I think τ_{i^*} is undefined. and suppose that for every $j \neq i^*$, granting or denying the current bidding interval to j does not change its probability of missing the deadline over the horizon **KM:** *this I don’t see, so that $\Delta_j^{\text{gross}}(s, t)$ is due only to changes in discounted reward timing. Then***

$$\Delta_{i^*}^{\text{gross}}(s, t) - \max_{j \neq i^*} \Delta_j^{\text{gross}}(s, t) \geq \delta M - 2r_{\max}.$$

KM: *One issue with this is that we promised earlier that unlike LSTF, our method is going to be “reward-aware.” But this Lemma suggests that our method is “deadline-aware,” isn’t it?*

By definition, the per-objective return can be written as

$$w_{i^*} = \gamma^{\tau_{i^*}} r_{i^*} \mathbf{1}[\tau_{i^*} < d_{i^*}] - M \mathbf{1}[\tau_{i^*} = \infty \text{ or } \tau_{i^*} \geq d_{i^*}].$$

Substituting this identity into the definition

$$\Delta_{i^*}^{\text{gross}}(s, t) = \tilde{V}_{i^*}^{\text{win}}(s, t) - \tilde{V}_{i^*}^{\text{lose}}(s, t) = \mathbb{E}[w_{i^*}(\Xi_{i^*}^{\text{win}}(s, t)) - w_{i^*}(\Xi_{i^*}^{\text{lose}}(s, t)) \mid s, t],$$

275 and grouping the reward and penalty parts gives

$$\begin{aligned} \Delta_{i^*}^{\text{gross}}(s, t) &= \mathbb{E}[\gamma^{\tau_{i^*}} r_{i^*} \mathbf{1}[\tau_{i^*} < d_{i^*}] \mid s, t, i^* \text{ wins at } (s, t)] - \mathbb{E}[\gamma^{\tau_{i^*}} r_{i^*} \mathbf{1}[\tau_{i^*} < d_{i^*}] \mid s, t, i^* \text{ loses at } (s, t)] \\ &\quad + M \left(\Pr[\tau_{i^*} = \infty \text{ or } \tau_{i^*} \geq d_{i^*} \mid s, t, i^* \text{ loses at } (s, t)] \right. \\ &\quad \left. - \Pr[\tau_{i^*} = \infty \text{ or } \tau_{i^*} \geq d_{i^*} \mid s, t, i^* \text{ wins at } (s, t)] \right). \end{aligned}$$

276 By assumption, the penalty term is at least δM . Moreover, since the truncated discounted reward
277 term $\gamma^{\tau_{i^*}} r_{i^*} \mathbf{1}[\tau_{i^*} < d_{i^*}]$ always lies in $[0, r_{\max}]$, the difference between the two reward expectations
278 for i^* is bounded below by $-r_{\max}$. Hence $\Delta_{i^*}^{\text{gross}}(s, t) \geq \delta M - r_{\max}$. For every $j \neq i^*$, the
279 no-expiry-change assumption **KM: make this assumption formal**. implies that $\Delta_j^{\text{gross}}(s, t)$ contains
280 only the discounted reward timing difference, and therefore $\Delta_j^{\text{gross}}(s, t) \leq r_{\max}$. Combining the two
281 bounds yields

$$\Delta_{i^*}^{\text{gross}}(s, t) - \max_{j \neq i^*} \Delta_j^{\text{gross}}(s, t) \geq M - C$$

282 for some constant $C \geq 0$, and moreover

$$M - C > \rho \quad \text{and} \quad \rho\beta > \max_{j \in [1:m]} \Delta_j^{\text{gross}}(s, t).$$

283 Then in any pure Nash equilibrium, policy i^* wins control.

284 *Proof.* By assumption,

$$\Delta_{i^*}^{\text{gross}}(s, t) - \max_{j \neq i^*} \Delta_j^{\text{gross}}(s, t) \geq M - C > \rho.$$

285 Hence the gap condition (??) of the previous theorem holds, and the claim follows. $\square$

286 The corollary should not be read as saying that M ought to be made arbitrarily large. Rather, it
287 identifies a relative scale separation: stronger missed-deadline penalties enlarge the regime in which
288 near-expiry targets are provably prioritized, provided the penalty signal dominates both the ordinary
289 reward scale $r_{\max}$ and the bidding increment scale ρ . Choosing M therefore remains a modeling
290 decision about how costly deadline misses should be in the underlying MO-MDP objective, not
291 merely a theorem-tuning parameter.

292 For the **All-Pay** result below, we analyze the standard continuous-bid approximation of the auction
293 game. **KM: Why do you say this?** This is justified when the bid grid is sufficiently fine and the
294 maximum bid is sufficiently large, so that the discrete bounded game approaches the continuous
295 all-pay contest.

296 The following theorem establishes that the **All-Pay** setting concentrates probability mass on the
297 RAUF choice whenever the top urgency is strictly ordered above the remaining urgencies.

298 **Theorem 3.** *In the All-Pay setting with $m \geq 3$, consider a single bidding round at the agent state*
299 *$s \in S$ and time t . Let i^* and j^* be policies for which the following gross gain order holds:*

$$\Delta_{i^*}^{\text{gross}} > \Delta_{j^*}^{\text{gross}} > \max_{k \notin \{i^*, j^*\}} \Delta_k^{\text{gross}}. \quad (2)$$

300 *Then there is no Nash equilibrium with pure strategies, although a unique Nash equilibrium exists*
301 *with mixed strategies (where at each (s, t) , each policy uses a probability distribution to select the*
302 *bid). In this Nash equilibrium, policy i^* wins control with probability $1 - \Delta_{j^*}^{\text{gross}} / 2\Delta_{i^*}^{\text{gross}}$.*

303 *Proof.* It follows from the work of Hillman and Riley ? that in the Nash equilibrium, the optimal bid
304 of the policies are as follows. The policy i^* picks a uniformly random bid b_i^* from $[0, \Delta_{j^*}^{\text{gross}} / \rho]$, the
305 policy j^* picks a uniformly random bid b_j^* from $(0, \Delta_{j^*}^{\text{gross}} / \rho]$ while picking the bid 0 with probability
306 $(\Delta_{i^*}^{\text{gross}} - \Delta_{j^*}^{\text{gross}}) / \Delta_{i^*}^{\text{gross}}$, and every other policy deterministically picks the bid 0. Clearly, if $b_i^* > b_j^*$,
307 then i^* wins control, and we need to show that $P[b_i^* > b_j^*] = 1 - \Delta_{j^*}^{\text{gross}} / 2\Delta_{i^*}^{\text{gross}}$. To show this, consider
308 the probability density function $f_{i^*}(b) = \rho / \Delta_{j^*}^{\text{gross}}$ of b_{i^*} , and the cumulative distribution function

309 $F_{j^*}(b) = (\Delta_{i^*}^{\text{gross}} - \Delta_{j^*}^{\text{gross}})/\Delta_{i^*}^{\text{gross}} + \rho b/\Delta_{i^*}^{\text{gross}}$ of b_{j^*} . Then,

$$\begin{aligned}
P[b_i^* > b_j^*] &= \int_{-\infty}^{\infty} F_{j^*}(b) f_{i^*}(b) db \\
&= \int_0^{\Delta_{j^*}^{\text{gross}}/\rho} \left(\frac{\Delta_{i^*}^{\text{gross}} - \Delta_{j^*}^{\text{gross}}}{\Delta_{i^*}^{\text{gross}}} + \frac{\rho b}{\Delta_{i^*}^{\text{gross}}} \right) \cdot \frac{\rho}{\Delta_{j^*}^{\text{gross}}} \cdot db \\
&= \int_0^{\Delta_{j^*}^{\text{gross}}/\rho} \frac{\Delta_{i^*}^{\text{gross}} - \Delta_{j^*}^{\text{gross}}}{\Delta_{i^*}^{\text{gross}}} \cdot \frac{\rho}{\Delta_{j^*}^{\text{gross}}} \cdot db + \int_0^{\Delta_{j^*}^{\text{gross}}/\rho} \frac{\rho b}{\Delta_{i^*}^{\text{gross}}} \frac{\rho}{\Delta_{j^*}^{\text{gross}}} \cdot db \\
&= \frac{\Delta_{i^*}^{\text{gross}} - \Delta_{j^*}^{\text{gross}}}{\Delta_{i^*}^{\text{gross}}} \cdot \frac{\rho}{\Delta_{j^*}^{\text{gross}}} \cdot \frac{\Delta_{j^*}^{\text{gross}}}{\rho} + \frac{\rho^2}{\Delta_{i^*}^{\text{gross}} \Delta_{j^*}^{\text{gross}}} \cdot \frac{(\Delta_{j^*}^{\text{gross}})^2}{2\rho^2} \\
&= \frac{\Delta_{i^*}^{\text{gross}} - \Delta_{j^*}^{\text{gross}}}{\Delta_{i^*}^{\text{gross}}} + \frac{\Delta_{j^*}^{\text{gross}}}{2\Delta_{i^*}^{\text{gross}}} \\
&= 1 - \frac{\Delta_{j^*}^{\text{gross}}}{\Delta_{i^*}^{\text{gross}}} + \frac{\Delta_{j^*}^{\text{gross}}}{2\Delta_{i^*}^{\text{gross}}} \\
&= 1 - \frac{\Delta_{j^*}^{\text{gross}}}{2\Delta_{i^*}^{\text{gross}}}
\end{aligned}$$

310

□

311 If the second inequality in (??) becomes an equality, then it is known that the Nash equilibrium
312 is no longer unique, and in fact, there will be a continuum of Nash equilibria ?, and reaching
313 the desirable equilibrium will require separate mechanisms. Instead, we argue that the event of
314 any of the inequalities in (??) becoming an equality is a measure zero event, and therefore it is
315 vanishingly unlikely in practice. Moreover, when $\Delta_{i^*}^{\text{gross}}$ is substantially larger than $\Delta_{j^*}^{\text{gross}}$, the
316 winning probability

$$1 - \frac{\Delta_{j^*}^{\text{gross}}}{2\Delta_{i^*}^{\text{gross}}}$$

317 approaches 1. Hence, although All-Pay need not deterministically implement RAUF, it still con-
318 centrates control on the most reward-aware urgent policy whenever the top-two urgency gap is
319 large.

320 **Global stochastic approximation guarantee.** The previous results are local one-step implementation
321 statements. We now state an assumption-based guarantee showing that, under stochastic target motion
322 and target appearance/disappearance, the auction policy can still achieve a constant fraction of the
323 completions attained by an optimal scheduler over a finite horizon.

324 Let π^{auc} denote the policy induced by a Nash equilibrium of the bidding game, let $p^{\text{NE}}(s, t)$ be
325 the distribution over winning objectives at bidding state (s, t) , and let $N^{\text{auc}}(T)$ and $N^*(T)$ be the
326 numbers of completed goals by horizon T under π^{auc} and the optimal scheduler, respectively. Writing
327 $u_i(s, t) := \Delta_i^{\text{gross}}(s, t)$, **KM: why define a new notation?** define

$$\begin{aligned}
u_{\max}(s, t) &:= \max_{i \in [1; m]} u_i(s, t), \\
H_{\lambda}(s, t) &:= \{i \in [1; m] \mid u_i(s, t) \geq \lambda u_{\max}(s, t)\}, \\
\beta(s, t) &:= \sum_{i \in H_{\lambda}(s, t)} p_i^{\text{NE}}(s, t),
\end{aligned}$$

328 where $\lambda \in (0, 1]$. **KM: so p_i^{NE} is the i -th component of the distribution?**

329 **Assumption 1** (Urgency-completion bridge). There exists $\kappa \in [0, 1]$ such that for every reachable
330 bidding state (s, t) and every objective i ,

$$q_i(s, t) \geq \kappa \frac{q_{\max}(s, t)}{u_{\max}(s, t)} u_i(s, t), \quad q_{\max}(s, t) := \max_{i \in [1; m]} q_i(s, t),$$

331 where $q_i(s, t)$ is the conditional probability that objective i is eventually completed by horizon T
332 **KM: But this could result in missing the deadline d_i , right?** if it is granted control at (s, t) .

Assumption 2 (Time Regularity). There exist constants $0 < \underline{\mu} \leq \bar{\mu} < \infty$ and $C_{\text{os}} \geq 0$ such that: (i) every attempt started under π^{auc} has conditional expected duration at most $\bar{\mu}$; (ii) no scheduler can achieve expected time per successful completion below $\underline{\mu}$; and (iii) the expected number of attempts started under π^{auc} before horizon T is at least $T/\bar{\mu} - C_{\text{os}}$.

Assumption 3 (Availability). There exists $q_0 \in (0, 1]$ such that at every reachable nonterminal bidding state (s, t) ,

$$q_{\text{max}}(s, t) \geq q_0.$$

Theorem 4 (Stochastic fraction-of-optimal completion guarantee). *Suppose the above three assumptions hold and $\beta(s, t) \geq \beta$. **KM: Please don't use the same symbol to denote a function as well as a constant uniformly over reachable bidding states for some $\beta \in (0, 1]$. **KM: Is it the same β as the maximum bid?** Then***

$$\mathbb{E}[N^{\text{auc}}(T)] \geq \lambda \eta \kappa q_0 \left(\frac{T}{\bar{\mu}} - C_{\text{os}} \right),$$

and moreover

$$\frac{\mathbb{E}[N^{\text{auc}}(T)]}{\mathbb{E}[N^*(T)]} \geq \lambda \eta \kappa q_0 \left(\frac{\underline{\mu}}{\bar{\mu}} - \frac{C_{\text{os}} \underline{\mu}}{T} \right).$$

Proof. At any reachable bidding state (s, t) , every objective in $H_\lambda(s, t)$ has urgency at least $\lambda u_{\text{max}}(s, t)$. Hence

$$\sum_{i=1}^m p_i^{\text{NE}}(s, t) u_i(s, t) \geq \lambda \eta(s, t) u_{\text{max}}(s, t) \geq \lambda \eta u_{\text{max}}(s, t).$$

By the urgency-completion bridge, the conditional probability that the auction-selected objective is eventually completed satisfies

$$\sum_{i=1}^m p_i^{\text{NE}}(s, t) q_i(s, t) \geq \kappa \frac{q_{\text{max}}(s, t)}{u_{\text{max}}(s, t)} \sum_{i=1}^m p_i^{\text{NE}}(s, t) u_i(s, t) \geq \lambda \eta \kappa q_{\text{max}}(s, t) \geq \lambda \eta \kappa q_0.$$

Summing this lower bound over the expected number of attempts started before horizon T and using the time-regularity assumption yields

$$\mathbb{E}[N^{\text{auc}}(T)] \geq \lambda \eta \kappa q_0 \left(\frac{T}{\bar{\mu}} - C_{\text{os}} \right).$$

On the other hand, the lower bound $\underline{\mu}$ on expected time per successful completion implies $\mathbb{E}[N^*(T)] \leq T/\underline{\mu}$. Combining the two inequalities gives the claimed ratio bound. $\square$

Remark 1 (Why λ can be chosen large in practice). The factor λ measures how close the selected “high-urgency” objectives are to the maximum reward-aware urgency. In deadline-driven environments, a natural choice is to define the urgent set through near-expiry targets. Suppose each target remains active for a lifetime L after appearing, let μ denote a typical completion-time scale, and call a target deadline-urgent when its remaining lifetime is at most $c\mu$ for some constant $c > 0$. When target ages are approximately asynchronous and spread across $[0, L]$, the fraction of such near-expiry targets is on the order of $\min\{1, c\mu/L\}$. For these targets, delaying service by even one bidding interval causes a substantial increase in expiry probability, so the missed-deadline term contributes on the order of M to their urgencies. By contrast, non-urgent targets have urgency variation dominated by the discounted reward scale r_{max} . Consequently, in regimes where $M \gg r_{\text{max}}$ and expiries are the main source of urgency, the objectives that matter for the approximation bound are not merely above average; they are typically close to the maximizer of $u_i(s, t)$, which makes choosing λ near 1 plausible.

Remark 2. Theorem ?? is stated for completion counts because this is the most robust quantity under stochastic target motion and dynamic target sets. When per-goal rewards are uniformly bounded away from zero and missed-goal penalties admit a common additive budget, the same theorem immediately yields a corresponding constant-factor guarantee for the total reward objective up to the associated reward-scaling and additive-shift terms.

Optional payoff corollary. For convenience, let $J^{\text{auc}}(T)$ and $J^*(T)$ denote the total payoff collected by horizon T under the auction policy and the optimal scheduler, respectively.

372 **Corollary 1** (Affine total-payoff guarantee). *Suppose the assumptions of Theorem ?? hold. Suppose*
 373 *moreover that there exist constants $0 < \underline{r} \leq \bar{r} < \infty$ and $B_{\text{miss}} \geq 0$ such that: (i) every completed*
 374 *objective contributes payoff in $[\underline{r}, \bar{r}]$; (ii) the total missed-deadline penalty incurred by the auction*
 375 *policy up to horizon T has expectation at most B_{miss} . Then*

$$\mathbb{E}[J^{\text{auc}}(T)] \geq \underline{r} \lambda \eta \kappa q_0 \left(\frac{T}{\underline{\mu}} - C_{\text{os}} \right) - B_{\text{miss}},$$

376 and

$$\mathbb{E}[J^{\text{auc}}(T)] \geq \frac{\underline{r}}{\bar{r}} \lambda \eta \kappa q_0 \left(\frac{\underline{\mu}}{\bar{\mu}} - \frac{C_{\text{os}} \underline{\mu}}{T} \right) \mathbb{E}[J^*(T)] - B_{\text{miss}}.$$

377 *Proof.* By assumption, the payoff contributed by completed objectives under the auction policy is at
 378 least $\underline{r} N^{\text{auc}}(T)$, while the expected missed-deadline penalty contributes at worst an additive loss of
 379 B_{miss} . Hence

$$\mathbb{E}[J^{\text{auc}}(T)] \geq \underline{r} \mathbb{E}[N^{\text{auc}}(T)] - B_{\text{miss}}.$$

380 Applying Theorem ?? yields the first inequality. For the second inequality, note that every completion
 381 under the optimal scheduler contributes at most $\bar{r}$, so

$$\mathbb{E}[J^*(T)] \leq \bar{r} \mathbb{E}[N^*(T)].$$

382 Combining this upper bound with the ratio bound from Theorem ?? gives the claim. $\square$

383 **GS: Something about $\beta = 1$ with WinnerPays?**

384 4 Learning Local Policies via Markov Games

385 We formalize our framework as an m -player general-sum Markov game (?), where each player
 386 optimizes one of the m objectives of the original MO-MDP. The game encodes the bidding mechanism
 387 in its transitions and rewards, and we therefore call it a *Markov bidding game*. Each state records the
 388 current MO-MDP state, the policy selected in the last bidding round, and the remaining time until the
 389 next bidding. At each state, players independently choose an action and a bid; transitions determine
 390 the winning policy based on the bids, while rewards capture the corresponding bidding penalties.

391 **Definition 1** (Markov bidding games). Let $\mathcal{M} = (S, A, T, R, \gamma, \mu)$ be an MO-MDP with m ob-
 392 jectives, τ be the bidding interval, β be the maximum bid, ρ be the bid penalty coefficient, and
 393 $\Delta \in \{\text{Winner-Pays}, \text{All-Pay}\}$ be the used penalty model. We define the m -player Markov bidding
 394 game as a tuple $\mathcal{G}_{\tau, \beta, \rho}^{\mathcal{M}} = (\hat{S}, \hat{A}, \hat{T}, \hat{R}, \gamma, \hat{\mu})$ where

- 395 • $\hat{S} = S \times [1; m] \times [0; \tau - 1]$ is the state space. An element of $\hat{S}$ is denoted (s, k, t) .
- 396 • $\hat{A} = (A \times \mathbb{Z}_{\geq 0})^m$ represents the joint extended action space of the m players, where each
 397 player independently selects an action from A and a nonnegative bid from $\mathbb{Z}_{\geq 0}$. We denote
 398 a set of actions chosen by the m players with $(\mathbf{a}, \mathbf{b}) = (a_i, b_i)_{i \in [1; m]} \in \hat{A}$.
- 399 • $\hat{T} : \hat{S} \times \hat{A} \times \hat{S} \rightarrow \mathbb{R}$ is the new transition function defined as,

$$\hat{T}((s, k, t), (\mathbf{a}, \mathbf{b}), (s', k', t')) := \begin{cases} T(s, a_{k'}, s') & t > 0 \wedge t' = t - 1 \wedge k' = k, \\ \frac{1}{|\mathbf{b}_{\max}|} T(s, a_{k'}, s') & t = 0 \wedge t' = \tau - 1 \wedge k' \in \mathbf{b}_{\max}, \\ 0 & \text{otherwise,} \end{cases}$$

400 where $\mathbf{b}_{\max} := \{i \mid b_i = \max\{b_1, \dots, b_m\}\}$ is the set of agents with the highest bid.

- 401 • $\{\hat{R}_i : \hat{S} \times \hat{A} \times \hat{S} \rightarrow \mathbb{R}\}_{i \in [1; m]}$ is the set of reward functions for the m agents with

$$\hat{R}_i((s, k, t), (\mathbf{a}, \mathbf{b}), (s', k', t')) := \begin{cases} R_i(s, a_{k'}, s') - \rho \cdot b_i & t = 0 \wedge (i = k' \vee \Delta = \text{All-Pay}), \\ R_i(s, a_{k'}, s') & \text{otherwise.} \end{cases}$$

• $\hat{\mu}$ is the initial state distribution over $\hat{S}$ defined as,

$$\hat{\mu}(s, k, t) := \begin{cases} \mu(s) & t = 0 \wedge k = i, \\ 0 & \text{otherwise,} \end{cases}$$

where $i \in [1; m]$ is an arbitrary agent.

In the Markov game $\mathcal{G}_{\tau, \beta, \rho}^{\mathcal{M}}$, each player i aims to compute its own selfish local policy of the form $\pi_i: (\hat{S} \times \hat{A})^* \times \hat{S} \mapsto (a_i, b_i) \in A \times \mathbb{Z}_{\geq 0}$, such that the discounted sum of its own reward is maximized, given that the other players are selfishly fulfilling their own objectives.

Learning policies in Markov bidding games. Since the transitions of the original MO-MDP are unknown, the transitions of the constructed Markov game in Definition ?? will also be unknown. Our goal is to learn the local policy of each player using RL. ? demonstrate that proximal policy optimization (PPO; ?) can be effective to train multiple policies concurrently for multi-agent settings. We follow their recipe and train all the local policies with PPO. Similar to ?, since the objectives in our environments share the same characteristics, we use the same actor and critic network for all the policies. At each iteration of PPO, we collect all the rollouts from each policy’s perspective and sample minibatches from this pool to perform the PPO updates.

Additionally, to cope with varying number of objectives and varying number of players in the resulting Markov game, we employ the *attention pooling* architecture to compress any number of objectives into a fixed-sized embedding. This design enables the agent to operate with an arbitrary number of policies (i.e., objectives) at deployment time. Each objective information vector is first processed by a fully connected neural network to produce a target embedding. These embeddings are then aggregated through a weighted sum, where the weights are computed by taking the dot product between each embedding and a learned query vector. This attention mechanism produces a fixed-dimensional representation that summarizes the set of objectives while allowing the model to adapt to varying objective counts.

5 Experimental Results

We conduct experiments to show the superior performance of our method compared to the baselines. We also perform ablations with respect to all the key components of the framework; and draw insights on their importance and how we can tune them. A detailed description of the environments and the baseline approaches is provided in Appendix ??, below we include a summary.

Environments. We evaluate our approach on two environments: a mobile cat feeder, introduced in Section ??, and the Atari game Assault. In the Cat Feeder environment, a robot moves on a 30×30 grid to feed up to m cats that move according to unknown stochastic processes. Each cat disappears after a fixed number of steps; feeding it yields a reward, while expiration incurs a penalty of equal magnitude. We train with $m = 7$ cats and test with up to 14. Each cat is served by a dedicated policy whose observation includes the robot position, the locations of active cats, and their remaining lifetimes.



Figure 2: Atari Assault env.

In Assault (Figure ??), the agent controls a missile launcher to destroy incoming alien ships while avoiding enemy missiles (?). We treat each ship as a separate objective and assign a local policy to pursue it, enabling a modular and interpretable design. Destroying a target yields reward, losing a life incurs a penalty, and performance is measured using the game score.

We also consider two variants of each environment: one without the attention pooling module (only for Cat Feeder; we do not use attention pooling for Atari Assault since there are only up to 3 targets) and another in which the local policies can only observe their own objectives (denoted Local Obs). Additionally, the supplementary material includes our codebase and recorded rollouts of our bidding-based policies on the two environments.

Baseline approaches. We compare against two baselines: a single PPO policy with aggregated rewards and deep W-learning (DWN) (?). The PPO baseline is implemented using CleanRL (?); for

Table 1: Performance (mean and standard deviation across 5 seeds) at the best evaluation iteration. Cat Feeder values measure performance as the difference between number of feeding requests completed and those that expired. Atari Assault values are the game score.

Method	Cat Feeder	Atari Assault
Winner-Pays	51.17 (4.49)	663.92 (23.73)
<i>w. Local Obs</i>	67.21 (5.75)	644.00 (28.31)
<i>w. No Attn. Pool.</i>	58.41 (5.09)	—
All-Pay	79.46 (3.68)	668.48 (21.34)
<i>w. Local Obs</i>	69.73 (3.57)	648.72 (53.85)
<i>w. No Attn. Pool.</i>	48.78 (47.10)	—
DWN	−2.28 (2.22)	310.80 (159.54)
Single-Policy PPO		430.08 (16.43)
<i>w. No Reward Shaping</i>	−48.75 (4.30)	—
<i>w. Nearest Shaping</i>	22.17 (36.34)	—
<i>w. Expiry Shaping</i>	−44.34 (1.38)	—

the Cat Feeder environment (with variable objective count) we additionally test sparse rewards and two shaping strategies. DWN trains one DQN policy per objective together with W-networks that estimate urgency scores of states. Hyperparameters are provided in ??, with Cat Feeder tuned using Optuna (?).

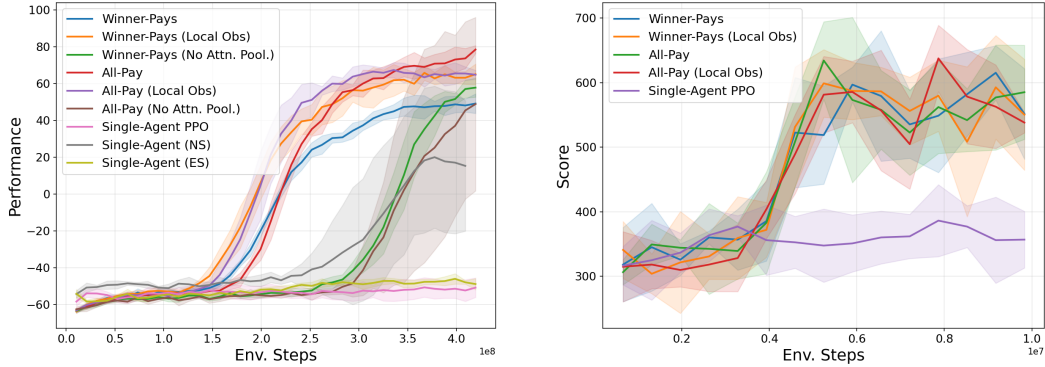


Figure 3: Learning curves for different methods on Cat Feeder (left) and Atari Assault (right). We plot mean over 5 seeds at each evaluation iteration and the shaded region is $\pm$ standard deviation.

Learning dynamics. ?? shows the learning curves for multi-policy bidding and single-policy PPO algorithms on the two environments. While the curves are more noisy with Atari Assault, they exhibit clearly distinguishable sigmoidal shapes with Cat Feeder. The curves for All-Pay and Winner-Pays methods start growing around the same time. Meanwhile, the variants with only local observations for each policy start growing much earlier. The variants with no attention pooling for targets only start growing much later. This validates the use of the attention pooling architecture to learn useful embeddings of targets. We remark that, based on the multi-agent PPO recipe of ?, we also found that using a small clip coefficient (~ 0.05) and large batch sizes was crucial to obtain slower but stable convergence of the policies towards a joint-equilibrium.

Comparison with baselines. ?? lists the performance of our method as well as the baselines on both environments. Our multi-policy methods outperform the baselines by a substantial margin. Across both the environments, we speculate that the poor performance of the single-policy PPO learner is due to the difficulty of credit assignment over a rollout in which there are multiple objectives that provide a positive reward. Additional reward shaping only provides moderate gains. In the Atari Assault environment, there is a delay between shooting a missile and it reaching its target. Since we only supply information from one frame in the observation, the learner could find it hard to associate the action of shooting and the reward from the death of a ship, especially if there is subsequent firing.

On the other hand, the credit assignment becomes easier when we separate the objectives and each learner only has to focus on one target.

DWN performs poorly in all experiments. We recall that this method learns a W-network trained to predict the negative TD-error for each policy when the policy does not play its action. However, in our environments, we have a long horizon and all the objectives are symmetric with similar Q-values. This means that the TD-error is small and does not provide a strong signal to choose between policies at most timesteps. In contrast, our approach allows the policies themselves to indicate the importance of playing an action.

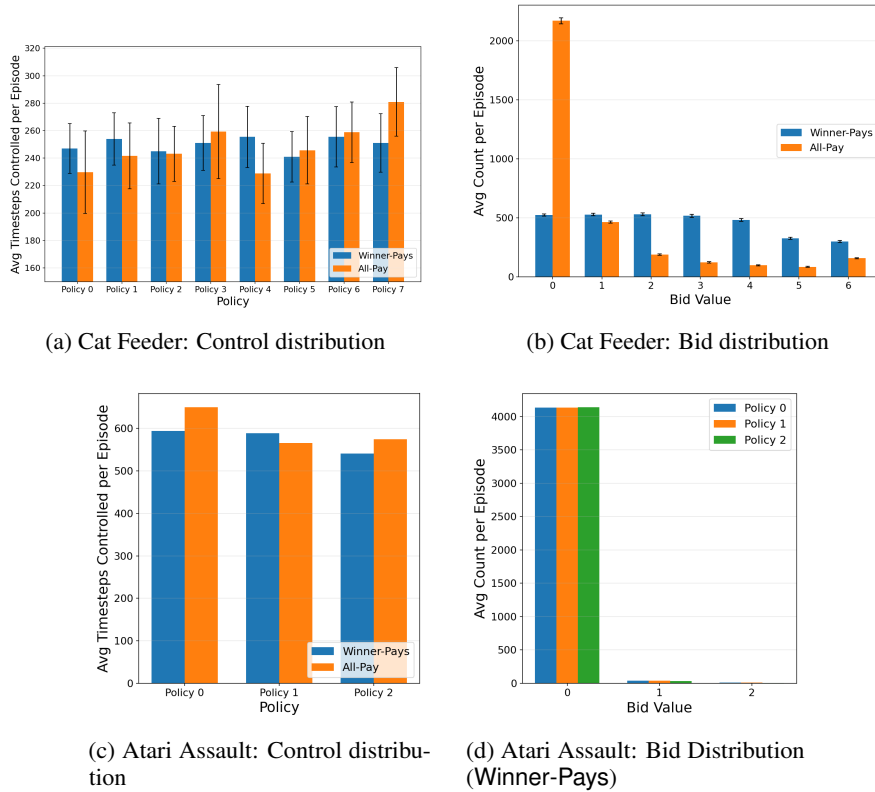


Figure 4: Comparison of control and bid distributions across environments and methods. The control distribution is the number of timesteps controlled by each policy, while the bid distribution is the distribution over bids picked by the policies.

Interpretability of policies and bidding behaviors. Video roll-outs show that bidding-based policies take control when they can accumulate reward within the action window, while each policy pursues only its own objective. ?? presents the distribution of control

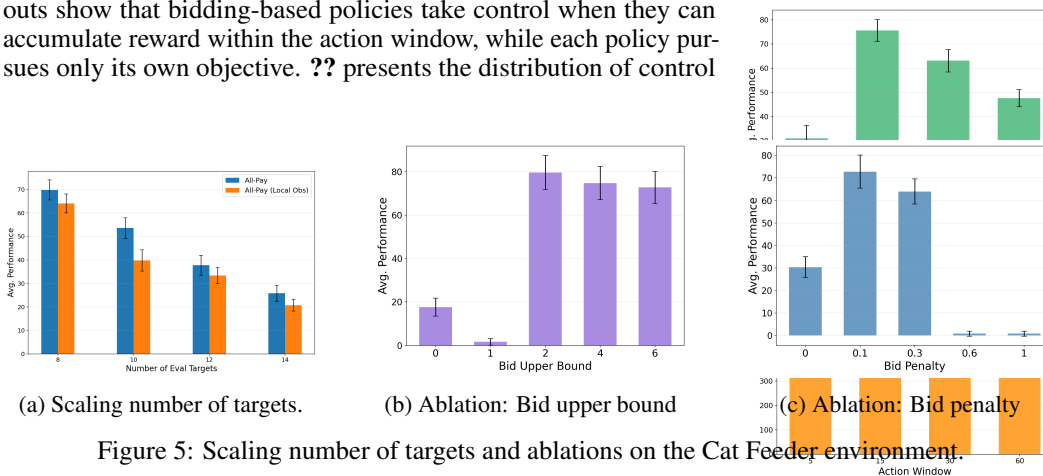


Figure 5: Scaling number of targets and ablations on the Cat Feeder environment.

Figure 6: Effect of action window on performance: Cat Feeder (top) and Atari Assault (bottom).

time and bids. Control is well distributed among policies. In Assault, bids are usually 0, indicating that random control assignment suffices since targets do not expire. In contrast, the Cat Feeder environment shows a wider bid spread (0–6). Here, All-Pay yields more conservative policies with mostly low bids, while Winner-Pays produces a more even distribution, reflecting greater willingness to bid high.

Scaling number of targets. We conduct an experiment with the Cat Feeder environment in which the number of cats seen during testing is larger than that during training. ?? shows that the policies continue to perform well up to 14 targets even though they were only trained to compete with up to 7 other policies. The attention pooling mechanism plays a critical role as it can generate useful embeddings of an arbitrary number of competing targets.

Further ablations. ?? shows that at least two bid levels are needed for good performance; with only one level, policies cannot compete effectively. ?? evaluates the bid penalty factor: without penalties all policies overbid, hurting performance, while overly large penalties discourage bidding. Finally, ?? studies the action window size, revealing an optimal value. Smaller windows cause excessive switching between policies, while larger ones prevent sufficient switching to serve all objectives.

6 Conclusion and Future Directions

We consider the problem of adapting RL policies due to evolving objectives, where any number of objectives may appear or disappear at runtime, and the agent must adapt as swiftly as possible. We present a compositional solution where each objective is served using a dedicated local policy, and each local policy competes against the other policies through an auction-based mechanism to select its preferred action. The training is formulated as a nonzerosum game between the local policies, and it is implemented using PPO in our framework. Using two challenging benchmarks, we demonstrate that our framework produces interpretable policies that achieve superior learning rate and dramatic performance improvement compared to the baseline approaches.

While we have only considered multi-objective environments in which the objectives belong to the same family, we will extend the framework to combinations of objective classes. The difficulty lies in the fact that a policy learned to compete against opponents from a given class may not work against opponents for a novel class of objectives seen at runtime. Additionally, richer bidding mechanisms might be necessary. For instance, while combining reachability and safety objectives, usually safety needs to get the priority when there is a tie in the bidding, instead of a random resolution.

References

- Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*, pages 2623–2631, 2019.
- Guy Avni, Thomas A Henzinger, and Ventsislav Chonev. Infinite-duration bidding games. *Journal of the ACM (JACM)*, 66(4):1–29, 2019.
- Guy Avni, Martin Kurečka, Kaushik Mallik, Petr Novotný, and Suman Sadhukhan. Bidding games on markov decision processes with quantitative reachability objectives. In *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems*, pages 161–169, 2025.
- Guy Avni, Kaushik Mallik, and Suman Sadhukhan. Auction-based scheduling. In *International Conference on Tools and Algorithms for the Construction and Analysis of Systems*, pages 153–172. Springer, 2024.
- Michael R Baye, Dan Kovenock, and Casper G De Vries. The all-pay auction with complete information. *Economic Theory*, 8(2):291–305, 1996.

535 M. G. Bellemare, Y. Naddaf, J. Veness, and M. Bowling. The arcade learning environment: An
536 evaluation platform for general agents. *Journal of Artificial Intelligence Research*, 47:253–279,
537 jun 2013.

538 Woohyeon Byeon, Giseung Park, Jongseong Chae, Amir Leshem, and Youngchul Sung. Multi-
539 objective reinforcement learning with max-min criterion: A game-theoretic approach. *arXiv*
540 *preprint arXiv:2510.20235*, 2025.

541 Quentin Delfosse, Jannis Blüml, Bjarne Gregori, Sebastian Sztiwrtnia, and Kristian Kersting.
542 Ocatari: Object-centric atari 2600 reinforcement learning environments, 2023.

543 Maria Fox, Alfonso Gerevini, Derek Long, Ivan Serina, et al. Plan stability: Replanning versus plan
544 repair. In *ICAPs*, volume 6, pages 212–221, 2006.

545 Saul Gass and Thomas Saaty. The computational algorithm for the parametric objective function.
546 *Naval research logistics quarterly*, 2(1-2):39–45, 1955.

547 Conor F Hayes, Roxana Rădulescu, Eugenio Bargiacchi, Johan Källström, Matthew Macfarlane,
548 Mathieu Reymond, Timothy Verstraeten, Luisa M Zintgraf, Richard Dazeley, Fredrik Heintz,
549 et al. A practical guide to multi-objective reinforcement learning and planning: Cf hayes et al.
550 *Autonomous Agents and Multi-Agent Systems*, 36(1):26, 2022.

551 Arye L Hillman and John G Riley. Politically contestable rents and transfers. *Economics & Politics*,
552 1(1):17–39, 1989.

553 Shengyi Huang, Rousslan Fernand Julien Dossa, Chang Ye, Jeff Braga, Dipam Chakraborty, Kinal
554 Mehta, and JoÃGo GM AraÃšjo. Cleanrl: High-quality single-file implementations of deep
555 reinforcement learning algorithms. *Journal of Machine Learning Research*, 23(274):1–18, 2022.

556 Mark Humphrys. W-learning: Competition among selfish q-learners. 1995.

557 Andrew J Lazarus, Daniel E Loeb, James G Propp, Walter R Stromquist, and Daniel H Ullman.
558 Combinatorial games under auction play. *Games and Economic Behavior*, 27(2):229–264, 1999.

559 Michael L Littman. Markov games as a framework for multi-agent reinforcement learning. In
560 *Machine learning proceedings 1994*, pages 157–163. Elsevier, 1994.

561 Beatriz M Méndez-Hernández, Erick D Rodríguez-Bazan, Yailen Martinez-Jimenez, Pieter Libin,
562 and Ann Nowé. A multi-objective reinforcement learning algorithm for jssp. In *International*
563 *Conference on Artificial Neural Networks*, pages 567–584. Springer, 2019.

564 Christos H. Papadimitriou. The euclidean traveling salesman problem is np-complete. *Theor. Comput.*
565 *Sci.*, 4(3):237–244, 1977.

566 Giseung Park, Woohyeon Byeon, Seongmin Kim, Elad Havakuk, Amir Leshem, and Youngchul Sung.
567 The max-min formulation of multi-objective reinforcement learning: From theory to a model-free
568 algorithm. *arXiv preprint arXiv:2406.07826*, 2024.

569 Matteo Pirota, Simone Parisi, and Marcello Restelli. Multi-objective reinforcement learning with
570 continuous pareto frontier approximation. In *Proceedings of the AAAI conference on artificial*
571 *intelligence*, volume 29, 2015.

572 Juan C Rosero, Nicolás Cardozo, and Ivana Dusparic. Multi-objective deep reinforcement learning
573 optimisation in autonomous systems. In *2024 IEEE International Conference on Autonomic*
574 *Computing and Self-Organizing Systems Companion (ACSOS-C)*, pages 97–102. IEEE, 2024.

575 Stuart J Russell and Andrew Zimdars. Q-decomposition for reinforcement learning agents. In
576 *Proceedings of the 20th international conference on machine learning (ICML-03)*, pages 656–663,
577 2003.

578 John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy
579 optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

580 Umer Siddique, Paul Weng, and Matthieu Zimmer. Learning fair policies in multi-objective (deep)
581 reinforcement learning with average and discounted rewards. In *International Conference on*
582 *Machine Learning*, pages 8905–8915. PMLR, 2020.

583 Kristof Van Moffaert, Madalina M Drugan, and Ann Nowé. Scalarized multi-objective reinforcement
584 learning: Novel design techniques. In *2013 IEEE symposium on adaptive dynamic programming*
585 *and reinforcement learning (ADPRL)*, pages 191–199. IEEE, 2013.

586 Kristof Van Moffaert and Ann Nowé. Multi-objective reinforcement learning using sets of pareto
587 dominating policies. *The Journal of Machine Learning Research*, 15(1):3483–3512, 2014.

588 Chao Yu, Akash Velu, Eugene Vinitsky, Jiaxuan Gao, Yu Wang, Alexandre Bayen, and Yi Wu. The
589 surprising effectiveness of ppo in cooperative multi-agent games. *Advances in neural information*
590 *processing systems*, 35:24611–24624, 2022.

591 A Experimental Setup (Detailed)

592 We consider two multi-objective environments for our empirical studies, namely a mobile cat feeder
593 and the Atari game called Assault. These environments are described below.

594 **Environment I: Cat Feeder.** The environment is as described in Section ?? and is modeled using
595 a 30×30 grid with up to m cats that are moving continuously according to unknown stochastic
596 processes. The value of m was set as 8 during the training time, whereas it was varied up to 14 during
597 the testing time. Each cat disappears after a given number of time steps. If the agent reaches a cat
598 before it disappears, it receives a reward, and otherwise it receives a penalty of equal magnitude. To
599 avoid sparsity of the reward function, we provide each policy a small reward for going closer to its
600 target. Performance is measured as the difference between the number of feeding requests completed
601 and the number of those that expired.

602 In our modular framework, each cat will be chased by a dedicated local policy. The observation
603 space of each local policy includes the current position of the robot, the current locations of the active
604 requests, and the time steps until the expiration of the requests.

605 **Environment II: Atari Assault.** We consider the Atari game called Assault, where the agent controls
606 a missile launcher to shoot oncoming alien ships while avoiding enemy missiles. The standard
607 approach is to view this as a single-objective environment, where the agent’s policy observes the
608 entire image or the 128 byte raw RAM state observations to determine the optimal actions (?).

609 For our modular framework, we break the objective down to a multi-objective problem, where each
610 alien ship is viewed as a separate objective, and each local policy attempts to destroy its assigned
611 target while avoiding enemy missiles. Not only does this facilitate an improvement in the performance,
612 as we will demonstrate, but also we obtain a transparent design, because which target is being chased
613 is revealed at each point. This modular design is made possible by the object-centric variant of the
614 Atari environment (?), which exposes the states of each individual target and the missiles by decoding
615 the RAM states. The full observation space of each local policy also include the current position of
616 the missile launcher, the current health, and the number of remaining lives. The local policy gets a
617 reward or a penalty upon destroying its target enemy ship or losing a life, respectively. Lastly, we use
618 the game score as the metric for evaluation.

619 **Baseline approaches.** We compare our method against two baselines: a single policy PPO with
620 weighted rewards and deep W-learning (DWN; ?). For the PPO baseline, since all the objectives
621 have the same importance across the two environments, we use the total reward across objectives at
622 each step as the reward function. We note that in the Atari Assault environment we do not use the
623 game score as a reward and instead use the same rewards perceived by the bidding-based policies to
624 ensure a fair comparison. The observation space remains the same as the multi-policy setting. In the
625 Cat Feeder environment, we implement three reward mechanisms to help guide learning: one only
626 with the sparse rewards from the environment, one with distance-based dense reward with respect to
627 the nearest feeding request (denoted Nearest Shaping/NS in the experiments), and finally one where
628 we provide distance-based reward with respect to the request that has the least amount of time until
629 expiration (denoted Expiry Shaping/ES). The PPO implementation is taken from CleanRL (?). We use
630 the specified PPO hyperparameters from CleanRL for Atari Assault and we tune the hyperparameters
631 for the Cat Feeder environment with 40 Optuna iterations (?).

DWN is another multi-policy method for multi-objective RL that uses DQN for each of the local policies and additionally learns a W-network for each policy that predict scalar weights. These weights are used to choose which policy’s action should be played at each step. The W-network is trained alongside the Q-network by updating it with a gradient step to predict the negative temporal difference (TD) error when the policy does not get to play its action. The assertion here is that the negative TD-error quantifies how important it is for the policy to play its action. The implementation is borrowed from the repository of ?. All hyperparameters for the baselines are listed in ??.

B Reproducibility Details

The code for all experiments is included in the supplementary materials.

All experiments use 5 random seeds: {1825, 410, 4507, 4013, 3658}.

B.1 Cat Feeder Environment

Parameter	Value
Grid size	30×30
Number of agents / targets	8
Target reward	50.0
Target expiry steps	200
Target expiry penalty	50.0
Max episode steps	2000
Moving targets	True
Direction change probability	0.1
Target move interval	5

Table 2: Cat Feeder shared (training) environment parameters.

B.2 Assault Environment

Parameter	Value
<i>Auction</i>	
Bid upper bound	6
Bid penalty	0.1
Action window	5
Distance reward scale	0.6
<i>Training</i>	
Iterations	400
Parallel environments	4096
Steps per rollout	256
Minibatches	256
PPO epochs	4
Learning rate	2.5×10^{-4} (annealed)
Discount γ	0.99
GAE λ	0.95
Clip coefficient	0.05
Entropy coefficient	0.03
Value function coefficient	1.0
Max gradient norm	0.5
<i>Network</i>	
Actor hidden layers	[128, 128, 128, 128]
Critic hidden layers	[256, 256, 256, 256]
Target attention pooling	True
Target embedding dim	64
Target encoder hidden layers	[64, 64]

Table 3: Cat Feeder multi-agent PPO hyperparameters (All-Pay and Winner-Pays).

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive” or “we were unable to find the license for the dataset we used”). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

Parameter	Value
<i>Training</i>	
Iterations	400
Parallel environments	4096
Steps per rollout	256
Minibatches	512
PPO epochs	8
Learning rate	1.74×10^{-4} (annealed)
Discount γ	0.963
GAE λ	0.970
Clip coefficient	0.327
Entropy coefficient	1.03×10^{-4}
Value function coefficient	1.076
Max gradient norm	0.840
Distance reward scale	0.0
Target attention pooling	False
<i>Network (shared with multi-agent PPO)</i>	
Actor hidden layers	[128, 128, 128, 128]
Critic hidden layers	[256, 256, 256, 256]

Table 4: Cat Feeder single-agent PPO baseline hyperparameters. Hyperparameters were tuned using Optuna with 40 trials. The nearest-target shaping variant is identical but uses a distance reward scale of 0.6 with shaping toward the nearest target.

Parameter	Value
<i>Training</i>	
Total timesteps	5×10^7
Parallel environments	256
Discount γ	0.99
Replay buffer size	1×10^6
Batch size	256
Learning starts	1×10^5
Train frequency	2560
W-network train delay	1×10^6
Target network update frequency	1000
Target network τ	1.0
<i>Networks</i>	
Q-network hidden layers	[256, 256, 256, 256]
W-network hidden layers	[128, 128, 128]
Q-network learning rate	1×10^{-4}
W-network learning rate	1×10^{-4}
<i>Epsilon schedules</i>	
Q ε start / min / decay	0.99 / 0.01 / 0.99
W ε start / min / decay	0.99 / 0.01 / 0.99

Table 5: Cat Feeder DWN baseline hyperparameters.

669 IMPORTANT, please:

- 670 • Delete this instruction block, but keep the section heading “NeurIPS Paper Checklist”,
- 671 • Keep the checklist subsection headings, questions/answers and guidelines below.
- 672 • Do not modify the questions and only use the provided macros for your answers.

673 1. Claims

Parameter	Value
Number of agents	3
Max enemies	3
Max episode steps	10000
Enemy destroy reward	10.0
Life loss penalty	10.0
Fire-while-hot penalty	8.0
Action window	15
Bid upper bound	2
Bid penalty	0.3

Table 6: Assault shared environment parameters.

Parameter	Value
<i>Training</i>	
Iterations	150
Parallel environments	128
Steps per rollout	512
Minibatches	8
PPO epochs	8
Learning rate	1×10^{-4} (annealed)
Discount γ	0.99
GAE λ	0.95
Clip coefficient	0.05
Entropy coefficient	0.05
Value function coefficient	0.5
Max gradient norm	0.5
Target KL	0.02
<i>Network</i>	
Actor hidden layers	[128, 128, 128, 128]
Critic hidden layers	[256, 256, 256, 256]

Table 7: Assault multi-agent PPO hyperparameters (All-Pay and Winner-Pays).

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [TODO]

Justification: [TODO]

Guidelines:

Parameter	Value
Learning rate	2.5×10^{-4} (annealed)
Clip coefficient	0.1
Entropy coefficient	0.01
Target KL	—

Table 8: Assault single-agent PPO baseline hyperparameters (differences from multi-agent PPO only; all other parameters are shared).

Parameter	Value
<i>Training</i>	
Total timesteps	1×10^7
Parallel environments	8
Discount γ	0.99
Replay buffer size	5×10^5
Batch size	256
Learning starts	1×10^4
Train frequency	80
W-network train delay	1×10^5
Target network update frequency	1000
Target network τ	1.0
<i>Networks</i>	
Q-network hidden layers	[256, 256, 256]
W-network hidden layers	[128, 128, 128]
Q-network learning rate	1×10^{-4}
W-network learning rate	1×10^{-4}
<i>Epsilon schedules</i>	
Q ε start / min / decay	0.99 / 0.01 / 0.995
W ε start / min / decay	0.99 / 0.01 / 0.995

Table 9: Assault DWN baseline hyperparameters.

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best

716 judgment and recognize that individual actions in favor of transparency play an impor-
717 tant role in developing norms that preserve the integrity of the community. Reviewers
718 will be specifically instructed to not penalize honesty concerning limitations.

719 3. Theory assumptions and proofs

720 Question: For each theoretical result, does the paper provide the full set of assumptions and
721 a complete (and correct) proof?

722 Answer: **[TODO]**

723 Justification: **[TODO]**

724 Guidelines:

- 725 • The answer [N/A] means that the paper does not include theoretical results.
- 726 • All the theorems, formulas, and proofs in the paper should be numbered and cross-
727 referenced.
- 728 • All assumptions should be clearly stated or referenced in the statement of any theorems.
- 729 • The proofs can either appear in the main paper or the supplemental material, but if
730 they appear in the supplemental material, the authors are encouraged to provide a short
731 proof sketch to provide intuition.
- 732 • Inversely, any informal proof provided in the core of the paper should be complemented
733 by formal proofs provided in appendix or supplemental material.
- 734 • Theorems and Lemmas that the proof relies upon should be properly referenced.

735 4. Experimental result reproducibility

736 Question: Does the paper fully disclose all the information needed to reproduce the main ex-
737 perimental results of the paper to the extent that it affects the main claims and/or conclusions
738 of the paper (regardless of whether the code and data are provided or not)?

739 Answer: **[TODO]**

740 Justification: **[TODO]**

741 Guidelines:

- 742 • The answer [N/A] means that the paper does not include experiments.
- 743 • If the paper includes experiments, a [No] answer to this question will not be perceived
744 well by the reviewers: Making the paper reproducible is important, regardless of
745 whether the code and data are provided or not.
- 746 • If the contribution is a dataset and/or model, the authors should describe the steps taken
747 to make their results reproducible or verifiable.
- 748 • Depending on the contribution, reproducibility can be accomplished in various ways.
749 For example, if the contribution is a novel architecture, describing the architecture fully
750 might suffice, or if the contribution is a specific model and empirical evaluation, it may
751 be necessary to either make it possible for others to replicate the model with the same
752 dataset, or provide access to the model. In general, releasing code and data is often
753 one good way to accomplish this, but reproducibility can also be provided via detailed
754 instructions for how to replicate the results, access to a hosted model (e.g., in the case
755 of a large language model), releasing of a model checkpoint, or other means that are
756 appropriate to the research performed.
- 757 • While NeurIPS does not require releasing code, the conference does require all submis-
758 sions to provide some reasonable avenue for reproducibility, which may depend on the
759 nature of the contribution. For example
 - 760 (a) If the contribution is primarily a new algorithm, the paper should make it clear how
761 to reproduce that algorithm.
 - 762 (b) If the contribution is primarily a new model architecture, the paper should describe
763 the architecture clearly and fully.
 - 764 (c) If the contribution is a new model (e.g., a large language model), then there should
765 either be a way to access this model for reproducing the results or a way to reproduce
766 the model (e.g., with an open-source dataset or instructions for how to construct
767 the dataset).

- (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so **[No]** is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The authors should answer **[Yes]** if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.

- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.

- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.

- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.

- 973 • We recognize that the procedures for this may vary significantly between institutions
974 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
975 guidelines for their institution.
976 • For initial submissions, do not include any information that would break anonymity (if
977 applicable), such as the institution conducting the review.

978 **16. Declaration of LLM usage**

979 Question: Does the paper describe the usage of LLMs if it is an important, original, or
980 non-standard component of the core methods in this research? Note that if the LLM is used
981 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
982 scientific rigor, or originality of the research, declaration is not required.

983 Answer: **[TODO]**

984 Justification: **[TODO]**

985 Guidelines:

- 986 • The answer [N/A] means that the core method development in this research does not
987 involve LLMs as any important, original, or non-standard components.
988 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
989 be described.